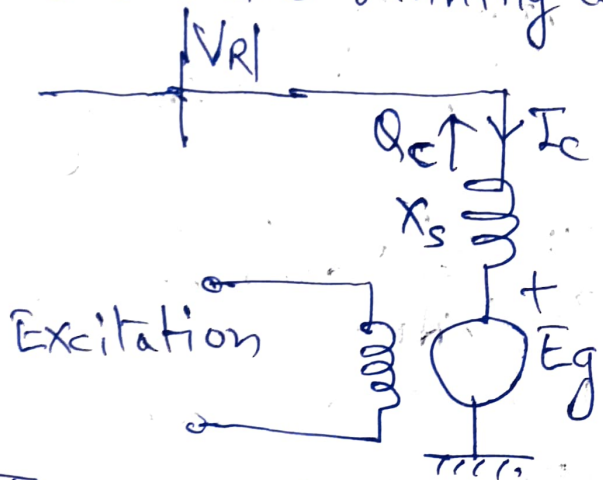


Synchronous condenser (also called Rotating VAR Generator)

Synchronous Motor running at no-load and having excitation adjustable over a wide range. It feeds positive VARs (leading) when overexcited (as capacitor) and feeds negative VARs (lagging) when under-excited (like inductor). A machine thus called Synchronous condenser.

Fig. shows a Synch. Motor connected to the receiving end bus and running at no-load.



Since m/c draws negligible real power from bus, so E_g & V_R are nearly in phase.
 $X_s \rightarrow$ synchronous reactance of motor
 If $|E_g| > |V_R|$ are in KV

$$\text{Then } I_c = \frac{|V_R| - |E_g|}{\sqrt{3} \times jX_s} \text{ KA}, \therefore jQ_c (\text{Reactive power}) = \frac{3|V_R| \angle 0^\circ}{\sqrt{3}} \angle -I_c$$

$$\text{OR } jQ_c = 3 \frac{|V_R|}{\sqrt{3}} \left[\frac{-|V_R| - |E_g|}{-jX_s \sqrt{3}} \right]$$

$$= j|V_R| (|E_g| - |V_R|) / X_s \text{ MVA}$$

$$Q_c = |V_R| (|E_g| - |V_R|) / X_s \text{ MVAR} \rightarrow \text{① (over excited)}$$

From ① If $|E_g| > |V_R|$ then M/c feeds $\oplus$ VARs into grid
 (ii) If $|E_g| < |V_R|$ " " " $\ominus$ " " " (under excited case)

Comparison

Synchronous Condenser

Static capacitors

- | | |
|---|---|
| 1. Provide both $\oplus$ and $\ominus$ Vars | only $\oplus$ Vars |
| 2. VARs can can be adjustable continuously | VARs can be changed in steps only |
| 3. Can supply large amount of VARs | Can supply small (limited) amount of VARs |
| 4. VAR injection is less sensitive to change in bus Voltage | as bus voltage reduces VAR supply also decreases. |